Blank Page

1. Beyond the Exponential Distribution

Let $Z \sim \text{Exponential}(\lambda)$, which means

$$f_Z(z) = \lambda \exp(-\lambda z) \text{ and } F_Z(z) = 1 - \exp(-\lambda z)$$

for $z \geq 0$.

1. Find the distribution of $X = \log Z$. → change of variables.
2. Use the probability integral transform to find a transformation from $U \sim \text{Uniform}[0,1]$ to X. → same as midterm 1, q 1

In class we considered

$$Z = -\log U$$

1)

$$F_X(x) = P(X \leq x)$$
$$= P(\log Z \leq x)$$
$$= P(Z \leq e^x)$$
$$= F_Z(e^x) = 1 - \exp(-\lambda e^x).$$

2) Invert $F_X(x) = U$

$$U = 1 - e^{-\lambda e^x}$$
$$1 - U = e^{-\lambda e^x}$$
$$\log(1-U) = -\lambda e^x$$
$$\frac{-1}{\lambda}\log(1-U) = e^x \Rightarrow X = \log\left[\frac{-1}{\lambda}\log(1-U)\right]$$

1. Beyond the Exponential Distribution

2. Chi Squared Acceptance Rejection

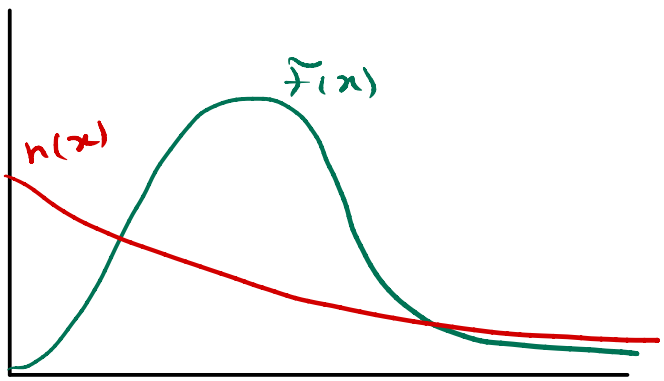
Let $X \sim \chi^2(4)$, which has PDF

$$f(x) = \frac{1}{4} x e^{-x/2}.$$

Use acceptance-rejection to simulate $\chi^2(4)$ random variables from the Exponential $(1/4)$ distribution, which has PDF

$$h(x) = \frac{1}{4} e^{-x/4}.$$

That is, for $f(x) = cg(x)h(x)$, find c, $g(x)$, and the probability of acceptance.



$$c = \sup_{x \geq 0} \frac{f(x)}{h(x)} = \sup_{x \geq 0} \left\{ \frac{1/4 \; x e^{-x/2}}{1/4 \; e^{-x/4}} \right\}$$

$$= \sup_{x \geq 0} \left\{ x e^{-x/4} \right\}$$

$$\frac{d}{dx} x e^{-x/4} = e^{-x/4} \cdot \frac{-1}{4} x e^{-x/4} = 0$$ [to find the critical point.]

2. Chi Squared Acceptance Rejection

$$e^{-x/4}\underbrace{\left[1-\frac{1}{4}x\right]}_{x=4}$$

$$c = 4e^{-1} = \frac{4}{e} > 1, \quad \text{prob(accept)} = \frac{e}{4} = \frac{2.71}{4}$$

$$g(x) = \frac{1}{c}\frac{\tilde{f}}{h} = \frac{1}{4/e}\, x e^{-x/4}$$

$$= \frac{1}{4} x e^{1-x/4}$$

3. Changing the Bias

Let $X_1, \ldots, X_n$ be an iid sample of data. Let $\hat{\theta} = \hat{\theta}(X_1, \ldots, X_n)$ be a biased estimator such that, for $n > 1$,

$$\mathrm{E}\hat{\theta} = \left(\frac{n^2 - 2}{n^2}\right)\theta.$$

Compute the bias of $\hat{\theta}$ and the bias of the jackknife estimator

$$\tilde{\theta} = n\hat{\theta} - (n-1)\left[\frac{1}{n}\sum_{i=1}^{n}\hat{\theta}_i\right].$$

Which bias is bigger in absolute value?

3. Changing the Bias

3. Changing the Bias

3. Changing the Bias